

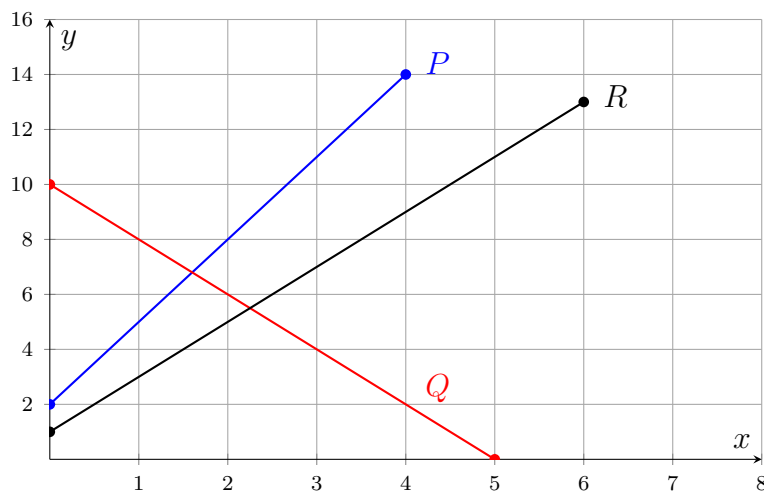
One Line, Four Ways to Show It

Contexts, tables, graphs, and equations that describe the same line

Directions

- Whatever form a relationship arrives in, look for the same two numbers: the starting value (when the input is 0) and the change per unit of input.
- Two representations match only when *both* numbers agree — one shared value is not enough.
- Write your answers as numbers with the units the story uses.

Graph Bank



Lines P, Q and R, each with two marked points. Only the problems that name a line by letter use this graph; every other problem supplies its own numbers.

1. A table of values is shown below.

x	0	1	2	3
y	6	10	14	18

- (a) What is the y -intercept of the line this table describes?

Answer: _____

- (b) What is its slope?

Answer: _____

2. A taxi charges \$3 to start the trip plus \$2 for each mile. Write the cost equation and use it to find the cost of a 12-mile ride, in dollars.

Answer: _____

3. Line P in the Graph Bank passes through the marked points $(0, 2)$ and $(4, 14)$.

- (a) Find the slope of line P .

Answer: _____

- (b) Write its equation and use it to find y when $x = 7$.

Answer: _____

4. The table below and the equation $y = 5x + 4$ are supposed to describe the same line.

x	0	2	4
y	4	14	24

- (a) What y -value does the table give at $x = 4$?

Answer: _____

- (b) What y -value does the equation give at $x = 4$?

Answer: _____

- (c) Do the table and the equation describe the same line? Explain how you decided.

Answer: _____

5. A phone battery starts a trip at 100 percent charge and drops 8 percentage points every hour. Write the charge equation and use it to find the charge after 9 hours.

Answer: _____

6. A tank is draining, as shown in the table.

Minutes	0	3	6	9
Litres	40	31	22	13

- (a) What is the y -intercept of the line this table describes?

Answer: _____

- (b) What is its slope, in litres per minute?

Answer: _____

7. Line Q in the Graph Bank passes through the marked points $(0, 10)$ and $(5, 0)$.

- (a) Find the slope of line Q .

Answer: _____

- (b) Write its equation and use it to find y when $x = 3$.

Answer: _____

8. A student claims that line R in the Graph Bank, which passes through the marked points $(0, 1)$ and $(6, 13)$, is described by the equation $y = 3x + 1$.

- (a) Find the slope of line R from its two marked points.

Answer: _____

- (b) What y -value does $y = 3x + 1$ give at $x = 6$?

Answer: _____

- (c) Does the equation describe line R ? Explain how you decided.

Answer: _____

9. A gym charges a \$50 joining fee plus \$18 for each month of membership, so the total cost after m months is $C = 18m + 50$ dollars.

- (a) Find the total cost after 7 months.

Answer: _____

- (b) After how many months does the total reach \$284?

Answer: _____

10. A table of values is shown below.

x	0	4	8	12
y	9	15	21	27

(a) By how much does y change from $x = 0$ to $x = 12$?

Answer: _____

(b) What is the slope of the line this table describes?

Answer: _____

11. A pool already holds 30 litres and gains 12 litres every minute. The table below and the equation $L = 12t + 30$ are both offered as descriptions.

Minutes	0	2	4
Litres	30	54	78

(a) How many litres does the table give at 4 minutes?

Answer: _____

(b) How many litres does the equation give at 4 minutes?

Answer: _____

(c) Explain why the story, the table and the equation all describe the same relationship.

Answer: _____

12. Line P in the Graph Bank, the table below, and the equation $y = 3x + 2$ all describe one relationship.

x	0	2	4
y	2	8	14

(a) What y -value does the table give at $x = 4$?

Answer: _____

(b) Use the equation to find y when $x = 20$.

Answer: _____

(c) Explain why the equation can answer part (b) when the graph and the table on their own cannot.

Answer: _____